

HIGHLIGHTS OF PRESCRIBING INFORMATION: METFORMIN HYDROCHLORIDE TABLETS

WARNING: LACTIC ACIDOSIS

Post-marketing cases of metformin-associated lactic acidosis have resulted in death, hypothermia, hypotension, and resistant bradyarrhythmias. The onset of metformin-associated lactic acidosis is often subtle, accompanied only by nonspecific symptoms such as malaise, myalgias, respiratory distress, somnolence, and abdominal pain. Risk factors include renal impairment, concomitant use of certain drugs (e.g., carbonic anhydrase inhibitors), age 65 years or older, radiological study with contrast, surgery and other procedures, hypoxic states, and excessive alcohol intake. If lactic acidosis is suspected, immediately discontinue metformin and institute general supportive measures in a hospital setting.

1 INDICATIONS AND USAGE

Metformin hydrochloride is a biguanide indicated as an adjunct to diet and exercise to improve glycemic control in adults and pediatric patients aged 10 years and older with type 2 diabetes mellitus.

2 DOSAGE AND ADMINISTRATION

- Adults: Recommended starting dose is 500 mg orally twice daily or 850 mg once daily with meals. Increase in increments of 500 mg weekly or 850 mg every 2 weeks up to maximum dose of 2550 mg per day in divided doses.
- Renal Impairment: Assess renal function prior to initiation:
 - eGFR 45 to 59 mL/min/1.73 m²: Initiation is not recommended, but if already taking, monitor renal function every 3 to 6 months.
 - eGFR 30 to 44 mL/min/1.73 m²: Maximum recommended dose is 1000 mg per day.
 - eGFR < 30 mL/min/1.73 m²: CONTRAINDICATED.
- Discontinue metformin at the time of or prior to iodinated contrast imaging procedures in patients with eGFR between 30 and 60 mL/min/1.73 m².

4 CONTRAINDICATIONS

- Severe renal impairment (eGFR below 30 mL/min/1.73 m²).
- Known hypersensitivity to metformin hydrochloride.
- Acute or chronic metabolic acidosis, including diabetic ketoacidosis, with or without coma.

5 WARNINGS AND PRECAUTIONS

- Lactic Acidosis: See Boxed Warning.
- Vitamin B12 Deficiency: Metformin may decrease vitamin B12 levels. Measure hematologic parameters annually and B12 at 2- to 3-year intervals.
- Hypoglycemia: Does not occur in patients receiving metformin alone under usual circumstances, but could occur in caloric deficiency, strenuous exercise, or combined with sulfonylureas or insulin.

6 ADVERSE REACTIONS

Most common adverse reactions (>5%): Diarrhea, nausea, vomiting, flatulence, asthenia, indigestion, abdominal discomfort, headache, and metallic taste in mouth. Adverse gastrointestinal events are most common during initiation and often transient.

7 DRUG INTERACTIONS

- Carbonic Anhydrase Inhibitors (e.g., Topiramate, Zonisamide): May increase risk